

Scaling Up the Experiment: Before and After

What changed when we moved from a tiny test set to a large one

To scale the experiment up to the full dataset, I split the work into three separate cloud jobs (I call them Kernel A, B, and C). This section explains what each one does and its current status. The rest of the report uses the results from A and B.

1 The three jobs (A, B, C)

Job	What it does	Status
Kernel A	Trains the Boston detector on the full data and measures the untouched pipeline once. This is the setup and timing check.	Done
Kernel B	Runs the main experiment: 15 retraining runs of the detector (varying data size, training length, and seed) and measures the effect on prediction and planning. This produces the headline result.	Done
Kernel C	Repeats the 15-style experiment with a class-balanced training set, to check whether balancing the rare classes adds further cases.	Running

So the numbers in this report come from the two finished jobs, A and B. Kernel C is a supporting check that is still running; its results will be added when it finishes. Kernel C is currently split across two accounts to fit inside the cloud time limit.

2 The dataset

The experiment now uses the full **nuScenes v1.0 trainval** dataset, the standard public benchmark for this kind of work. It contains 850 scenes: 467 recorded in Boston and 383 in Singapore. We train the detector on Boston and fine-tune it on Singapore, so the retraining is a real geographic domain shift. The dataset (about 45 GB) was already hosted on the cloud platform, so no download or extra compute cost was needed.

The earlier version used only nuScenes-mini, a tiny 10-scene demo subset. That is the core difference: we moved from the demo subset to the full benchmark.

3 The one change that drives everything

The previous version of the experiment used a very small test set: only 3 scenes, which is about 121 camera frames. The new version uses a large test set: 50 scenes, which is 2,020 frames. A scene is a 20-second driving clip and holds roughly 40 frames, so the frame count is the fair way to compare the two.

One point to be clear about: we train on all the Singapore data (229 scenes), but we evaluate on 50 of the 154 Singapore validation scenes, chosen evenly across the set. This keeps the running

time manageable across 15 retraining runs while still giving a test set that is many times larger than before. The 50 scenes are fixed, so every run is measured on exactly the same frames.

Nothing about the method changed. Only the amount of data changed. Every result below follows from that.

4 How much data, before and after

	Before (mini)	After (full trainval)
Boston training scenes	6 scenes	280 scenes, 11,087 images
Singapore fine-tune data	119 images	229 scenes
Test set	3 scenes, 121 frames	50 scenes, 2,020 frames

The test set grew from 121 frames to 2,020 frames, roughly 17 times larger.

5 The pipeline (unchanged)

The system is the same three-part self-driving pipeline:

- **C1, Perception:** a camera detector (YOLOv11) that finds cars, pedestrians, and other agents.
- **C2, Prediction:** predicts where those agents will move.
- **C3, Planning:** decides the car’s driving path.

We retrain only C1 on new-city data and measure what happens to C2 and C3. That part did not change.

6 The real numbers, side by side

Both tables below report the same reference update: the detector is trained on Boston, then fine-tuned on Singapore, and every component is measured on the Singapore test set. “Isolated” means the component is fed perfect ground-truth inputs; “pipeline” means it is fed the live output of the part above it. The gap between the two is the cascade.

The last column reads each row as **better** (green sense: score moved the good way) or **WORSE** (bad way). Entangled enhancement is the pattern where C1 is better, C2 is better, and yet C3 is WORSE at the same time.

Before: mini test set (3 scenes, 121 frames)

Pattern here: **C1 WORSE**, C2 better, **C3 WORSE**. Because the detector itself got worse, this is *not* entangled enhancement (that needs C1 to get better). This is why the mini paper had to add a class-balancing step to force C1 up.

After: full test set (50 scenes, 2,020 frames)

Pattern here: **C1 better**, **C2 better**, **C3 WORSE**. All three at once, which is exactly **entangled enhancement**, and it appears with no trick. The small-data drop in C1 from the mini set is gone.

Part / metric	Before	After	$\Delta\%$	Direction
C1 detector mAP@50	0.1185	0.0644	-45.7	WORSE
C2 pred., isolated	2.09	2.09	0.0	(frozen)
C2 pred., pipeline	15.19	3.95	-74.0	better
C3 plan, isolated	4.93	4.93	0.0	(frozen)
C3 plan, pipeline	6.30	6.58	+4.5	WORSE
C3 collision rate	0.00	0.00	0.0	none

Part / metric	Before	After	$\Delta\%$	Direction
C1 detector mAP@50	0.137	0.256	+86.6	better
C2 pred., isolated	1.12	1.12	0.0	(frozen)
C2 pred., pipeline	7.93	5.76	-27.4	better
C3 plan, isolated	4.64	4.64	0.0	(frozen)
C3 plan, pipeline	5.55	5.67	+2.2	WORSE
C3 collision rate	0.02	0.02	0.0	none

7 The full result: all 15 retraining runs (after)

This is the complete Kernel B result. Each row is one retraining of the detector, under a different setting. The first three columns are the settings we varied: **Data** is how much Singapore data we fine-tuned on (full, half, or a quarter), **Ep.** is how many training epochs, and **Seed** is the random seed (repeating each setting rules out luck).

The three “ Δ ” columns show whether each part improved or worsened compared to the baseline: for C1 a **positive** Δ is better (higher detection score); for C2 and C3 a **negative** Δ is better (lower error). The last column marks the row **YES** when C1 improved and C3 worsened at the same time, which is entangled enhancement. Those rows are in bold.

Baseline (before any retraining): C1 mAP = 0.137, C2 pipeline = 7.93 m, C3 pipeline = 5.55 m.

How to read it in one look: the C1 column is positive in every row (detector always improved), the C2 column is negative in every row (prediction always improved), and the C3 column is positive in 6 rows (planner worse). Those 6 rows are the entangled-enhancement cases: a better detector, a better predictor, and yet a worse driving plan.

8 Summary of what changed

The plan-shift range also became tighter, from 1.33–7.80 m down to 0.78–1.50 m. This is expected: with a much larger test set the measurement is less noisy, so the extreme values from the tiny 3-scene set do not appear.

9 Baseline system, before any retraining

For reference, the untouched pipeline measured on the 2,020-frame test set:

The planner error in the pipeline (5.55 m) is larger than the planner error in isolation (4.64 m). That gap is the cascade: the planner inherits the error that flows down from perception. It now holds across 50 scenes instead of 3.

Data	Ep.	Seed	C1 Δ (+ better)	C2 Δ (m) (− better)	C3 Δ (m) (− better)	Ent. enh.?
Full	10	1	+0.119	−2.17	+0.124	YES
Full	10	2	+0.122	−2.28	+0.193	YES
Full	10	3	+0.129	−2.15	+0.187	YES
Half	10	1	+0.114	−2.55	−0.121	no
Half	10	2	+0.120	−2.71	−0.230	no
Half	10	3	+0.106	−1.77	−0.151	no
Qtr	10	1	+0.121	−2.71	+0.073	YES
Qtr	10	2	+0.113	−1.29	−0.093	no
Qtr	10	3	+0.101	−2.82	+0.008	YES
Full	5	1	+0.138	−1.08	−0.095	no
Full	5	2	+0.122	−1.88	−0.130	no
Full	5	3	+0.130	−1.08	−0.041	no
Full	20	1	+0.142	−2.59	+0.037	YES
Full	20	2	+0.126	−1.77	−0.083	no
Full	20	3	+0.130	−2.44	−0.020	no
Totals			15/15 better	15/15 better	6/15 worse	6 YES

What we measured	Before (3 sc., 121 fr.)	After (50 sc., 2,020 fr.)
Detector improves after retraining?	0 of 15	15 of 15
Prediction improves?	mixed	15 of 15
Entangled enhancement seen directly?	0 of 15 (needed a trick)	6 of 15 (no trick)
Collisions happen?	Never (0)	Yes (2%)
Plan shift range	1.33 to 7.80 m	0.78 to 1.50 m

Metric	Value	Reading
C1 detector mAP	0.137	detector works
C2 prediction, isolated	1.12 m	clean prediction
C2 prediction, pipeline	7.93 m	error inherited from the detector
C3 planning, isolated	4.64 m	planner alone
C3 planning, pipeline	5.55 m	worse than isolated
C3 collision rate	2%	collisions now occur

10 What this means for the paper

1. The main claim is stronger. Entangled enhancement in a real driving pipeline is now shown by 6 ordinary retraining runs on full data, rather than by a special class-balancing step on 3 scenes.
2. One old finding is removed. The “detector score drops after retraining” result was caused by the small 119-image fine-tune set. With full data it does not happen, so that finding is

dropped.

3. A new result appears. Collisions occur at full scale (they never did on 3 scenes), which makes the safety part of the proposed method (CARA) testable.

11 Still coming

Kernel C (the class-balanced runs) is still running, split across two accounts to fit the cloud time limit. In the old small-scale experiment, this class-balancing step was necessary to obtain the headline result. At full scale it is no longer necessary, because Kernel B already produces the result on its own. Kernel C now serves only as a supporting check on whether balancing the rare classes adds more entangled-enhancement cases. Its numbers will be added when it finishes.